Miao Lin

PH.D. CANDIDATE · DEPARTMENT OF COMPUTER SCIENCE

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Education

Old Dominion University *Norfolk, US*
PhD, Department of Computer Science *08/2023 - Present*

Old Dominion University *Norfolk, US*
Master of Science *08/2023 - 12/2025*

Central South University *Changsha, China*
Master of Medicine *08/2020 - 06/2023*

Research Experience

Trustworthy AI
My current research secures AI models for edge devices by advancing defense mechanisms against emerging adversarial and data-poisoning/backdoor threats. Over the past two years, I identified a critical vulnerability and, leveraging this insight, designed a novel defense with rigorous theoretical analysis. The approach shows promise for real-world applications such as voice assistants and facial recognition, and I am now refining and expanding the algorithm to improve robustness and adaptability across broader deployments.

Teaching & Outreach

01/2026 - Present — **Teaching Assistant**, School of Cybersecurity, ODU. Designed and developed Capture-the-Flag (CTF) challenges exploring security vulnerabilities in large language models (LLMs) and multi-modal language models (MLLMs).

04/2026 - 05/2026 — **Guest Lecturer**, *CYSE615 (Mobile and Wireless Security)*, School of Cybersecurity, ODU. Presented an overview of backdoor attacks in CNNs and LLMs, highlighting how hidden triggers can compromise model behavior after deployment. Discussed the broader security implications of AI backdoors for mobile, wireless, and edge-computing environments.

09/2025 - 10/2025 — **Guest Lecturer**, *CYSE635 (AI Security & Privacy)*, School of Cybersecurity, ODU. Taught core concepts of cybersecurity defense, including common software vulnerabilities, attack vectors, and mitigation best practices. Conducted hands-on lab sessions using SEED Labs, working with real-world malware and exploits to reinforce defensive techniques.

05/2025 - 10/2025 — **Entrepreneur Lead**, *NSF Regional I-Corps Program*. Conducted 40+ customer-discovery interviews on privacy-tech needs; refined problem–solution fit, value proposition, and go-to-market strategy for a cybersecurity startup concept.

07/2023 - 12/2024 — **Teaching Assistant**, *CS 467/567: Introduction to Reverse Software Engineering*, Department of Computer Science, ODU. Led labs and grading; supported lectures on static/dynamic analysis, Win x86/64, API hooking, DLL/process injection, and network analysis; mentored an AI-assisted malware-analysis capstone.

Academic Service

- **Reviewer**: ICNC, Journal of Information Security and Applications. *2026*
- **Curriculum Co-designer**, CYSE635/CS695: AI Security & Privacy, ODU. *07/2024–05/2025*

- **Curriculum Developer**, NSF T3-CIDERS Cyber Training Program. *05/2024–08/2024*

Publications

M. Lin, M. D. Mazumder, F. Yu, L. Li, D. Takabi, R. Ning. Laplace-Bridged Randomized Smoothing for Fast Certified Robustness. *arXiv preprint arXiv:2604.24993, 2026.*

M. Lin, F. Yu, R. Ning, L. Li, Q. Lou, M. Zheng, C. Xin, H. Wu. RPP: a certified poisoned-sample detection framework for backdoor attacks under dataset imbalance. *arXiv preprint arXiv:2602.00183, 2026.*

M. Lin, J. Zhang, J. Li, F. Yu, L. Li, C. Xin, H. Wu, R. Ning. ACS-Boot: efficient randomized smoothing for robustness certification on resource-constrained edge devices. In *IEEE Conference on Computer Communications* (INFOCOM), 2026.

Z. Chen, J. Li, **M. Lin**, A. Mao, L. Li, R. Ning, C. Xin, H. Wu. TrojanEdge: mutual information–enhanced robust and persistent backdoor attacks for edge and on-device deployments. In *IEEE Conference on Computer Communications* (INFOCOM), 2026.

D. Xiong#, **M. Lin**#, D. Lang, *et al.* SLCO4A1-mediated transmembrane transport of lysionotin attenuates acute lung injury by activating the AMPK/Nrf2 signaling pathway. *Phytotherapy Research*. 2025; 39(10):4404–4427.

W. Xie, L. Deng, **M. Lin**, *et al.* Sirtuin 1 mediates the protective effects of echinacoside against sepsis-induced acute lung injury by regulating the NOX4–Nrf2 axis. *Antioxidants*. 2023; 12(11):1925.

M. Lin, W. Xie, D. Xiong, *et al.* Cyasterone ameliorates sepsis-related acute lung injury via AKT(Ser473)/GSK3β(Ser9)/Nrf2 pathway. *Chinese Medicine*. 2023; 18:136.

Awards & Honors

- **Student Travel Awards, IEEE INFOCOM** *2026*
- **2024-2025 Dominion Scholar, ODU** *2024*
- **Second Prize Scholarship, CSU** *2020, 2021, 2022*
- **Provincial Excellent Undergraduate** *2020*
- **National Scholarship, CSU** *2019*